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Phase-change memory devices

Abstract

In an embodiment, a device includes: a first metallization layer over a substrate, the substrate including active devices; a first bit line over the first metallization layer, the first bit line connected to first interconnects of the first metallization layer, the first bit line extending in a first direction, the first direction parallel to gates of the active devices; a first phase-change random access memory (PCRAM) cell over the first bit line; a word line over the first PCRAM cell, the word line extending in a second direction, the second direction perpendicular to the gates of the active devices; and a second metallization layer over the word line, the word line connected to second interconnects of the second metallization layer.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 17/867,460, filed on Jul. 18, 2022, entitled “Phase-Change Memory Device and Method,” which is a divisional of U.S. patent application Ser. No. 17/072,897, filed on Oct. 16, 2020, entitled “Phase-Change Memory Device and Method,” now U.S. Pat. No. 11,411,181, issued on Aug. 9, 2022, which claims the benefit of U.S. Provisional Application No. 63/001,944, filed on Mar. 30, 2020, which applications are hereby incorporated herein by reference.

BACKGROUND

(1) Semiconductor memories are used in integrated circuits for electronic applications, including radios, televisions, cell phones, and personal computing devices, as examples. One type of semiconductor memory is phase-change random access memory (PCRAM), which involves storing values in phase change materials, such as chalcogenide materials. Phase change materials can be switched between an amorphous phase (in which they have a low resistivity) and a crystalline phase (in which they have a high resistivity) to indicate bit codes. A PCRAM cell typically includes a phase change material (PCM) element between two electrodes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a block diagram of a semiconductor device, in accordance with some embodiments.

(3) FIG. 2 is a cross-sectional view of a semiconductor device, in accordance with some embodiments.

(4) FIGS. 3 through 21B are various views of intermediate stages in the manufacturing of a semiconductor device, in accordance with some embodiments.

(5) FIG. 22 is a cross-sectional view of a semiconductor device, in accordance with some other embodiments.

(6) FIGS. 23A through 23E are three-dimensional views of intermediate stages in a self-aligned patterning process for forming PCRAM cells, in accordance with some embodiments.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature

in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) In accordance with some embodiments, PCRAM cells are formed in a self-aligned manner with multiple patterning processes. The PCRAM cells can thus be formed at a smaller pitch and with smaller critical dimensions. The performance and density of the PCRAM cells may thus be improved.

(10) FIG. 1 is a block diagram of a semiconductor device 50, in accordance with some embodiments. The semiconductor device 50 includes a PCRAM array 52, a row decoder 54, and a column decoder 56. The PCRAM array 52 includes PCRAM cells 58 arranged in rows and columns. The row decoder 54 may be, e.g., a static CMOS decoder, a pseudo-NMOS decoder, or the like. During operation, the row decoder 54 selects desired PCRAM cells 58 in a row of the PCRAM array 52 by activating the respective word line 62 for the row. The column decoder 56 may be, e.g., a static CMOS decoder, a pseudo-NMOS decoder, or the like, and may include writer drivers, sense amplifiers, combinations thereof, or the like. During operation, the column decoder 56 selects bit lines 66 for the desired PCRAM cells 58 from columns of the PCRAM array 52 in the selected row, and reads data from or writes data to the selected PCRAM cells 58 with the bit lines 66.

(11) Although embodiments herein are described in the context of PCRAMs, it should be appreciated that similar techniques could be applied in other memories that use programmable resistance elements. For example, similar techniques could be used to manufacture magnetoresistive random-access memories (MRAMs), resistive random access memories (RRAMs), memories with a selector structure, and the like.

(12) FIG. 2 is a cross-sectional view of the semiconductor device 50, in accordance with some embodiments. FIG. 2 is a simplified view, and some features of the semiconductor device 50 (discussed below) are omitted for clarity of illustration. The semiconductor device 50 includes a logic region 50L and a memory region 50M. Memory devices (e.g., PCRAMs) are formed in the memory region 50M and logic devices (e.g., logic circuits) are formed in the logic region 50L. For example, the PCRAM array 52 (see FIG. 1) can be formed in the memory region 50M, and the row decoder 54 and the column decoder 56 (see FIG. 1) can be formed in the logic region 50L. The logic region 50L may occupy most of the area of the semiconductor device 50. For example, the logic region 50L may occupy from 95% to 99% of the area of the semiconductor device 50, with the memory region 50M occupying the remaining area of the semiconductor device 50. The memory region 50M can be disposed at an edge of the logic region 50L, or the logic region 50L can surround the memory region 50M.

(13) The logic region 50L and memory region 50M are formed over a same substrate, e.g., a semiconductor substrate 70. The semiconductor substrate 70 may be silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate 70 may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, gallium nitride, indium phosphide,

indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multilayered or gradient substrates, may also be used.

(14) Devices **72** are formed at the active surface of the semiconductor substrate **70** (e.g., the surface facing upwards in FIG. 2). The devices **72** may be active devices or passive devices. For example, the electrical components may be transistors, diodes, capacitors, resistors, or the like, formed by any suitable formation method. The devices **72** are interconnected to form the memory devices and logic devices of the semiconductor device **50**. For example, some of the devices **72** may be access transistors for the PCRAM cells **58**.

(15) One or more inter-layer dielectric (ILD) layer(s) **74** are formed on the semiconductor substrate **70**, and electrically conductive features, such as contact plugs **76**, are formed physically and electrically coupled to the devices **72**. The ILD layer(s) **74** may be formed of any suitable dielectric material, for example, an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; a nitride such as silicon nitride; or the like. The ILD layer(s) may be formed by any suitable deposition process, such as spin coating, physical vapor deposition (PVD), chemical vapor deposition (CVD), the like, or a combination thereof. The electrically conductive features in the ILD layer(s) may be formed through any suitable process, such as deposition, damascene (e.g., single damascene, dual damascene, etc.), the like, or combinations thereof.

(16) An interconnect structure **78** is formed over the semiconductor substrate **70**, e.g., over the ILD layer(s) **74**. The interconnect structure **78** interconnects the devices **72** to form integrated circuits in each of the logic region **50L** and memory region **50M**. The interconnect structure **78** includes multiple metallization layers **M1-M6**. Although six metallization layers are illustrated, it should be appreciated that more or less metallization layers may be included. Each of the metallization layers **M1-M6** includes metallization patterns in dielectric layers. The metallization patterns are electrically coupled to the devices **72** of the semiconductor substrate **70**, and include, respectively, metal lines **L1-L6** and vias **V1-V6** formed in one or more inter-metal dielectric (IMD) layers. The interconnect structure **78** may be formed by a damascene process, such as a single damascene process, a dual damascene process, or the like. In some embodiments, the contact plugs **76** are also part of the metallization patterns, such as part of the lowest layer of metal vias **V1**.

(17) The PCRAM cells **58** of the PCRAM array **52** (see FIG. 1) are formed in the interconnect structure **78**. The PCRAM cells **58** can be formed in any of the metallization layers **M1-M6**, and are illustrated as being formed in an intermediate metallization layer **M5**. Each PCRAM cell **58** includes a bottom electrode **82**, a PCM element **84** on the bottom electrode **82**, and a top electrode **86** on the PCM element **84**. Word lines **62** extend along respective rows of the PCRAM cells **58** and are connected to the top electrodes **86** of the respective rows of the PCRAM cells **58**. Bit lines **66** extend along respective columns of the PCRAM cells **58** and are connected to the bottom electrodes **82** of the respective columns of the PCRAM cells **58**. One or more additional IMD layer(s) **88** can be formed around the PCRAM cells **58**. The IMD layer(s) **88** surround and protect the components of the PCRAM cells **58**. The resistance of a PCM element **84** is programmable, and can be changed between a high resistance ($R_{sub,AP}$), which can signify a code such as a “1,” and a low resistance ($R_{sub,P}$), which can signify a code such as a “0.” As such, a code can be written to a PCRAM cell **58** by programming the resistance of its PCM element **84** with its corresponding access transistor, and a code can be read from a PCRAM cell **58** by measuring the resistance of its PCM element **84** with its corresponding access transistor.

(18) The PCRAM cells **58** are electrically coupled to the devices **72**. The bit lines **66** are connected to conductive features (e.g., interconnects) of an underlying metallization pattern, such as to the metallization layer **M4** in the illustrated example, by conductive vias **92**. The word lines **62** are connected to conductive features (e.g., interconnects) of an overlying metallization pattern, such as to the metallization layer **M6** in the illustrated example, by conductive vias **94**. A first subset of the

devices **72** (e.g., access transistors), such as devices of the row decoder **54**, are electrically coupled to the word lines **62**. The bit lines **66** are electrically coupled to a second subset of the devices **72**, such as devices of the column decoder **56**.

(19) Referring initially to FIG. **21B**, a simplified top-down view of a portion of the memory region **50M** is shown. Some features of the semiconductor device **50** (discussed in greater detail below) are omitted for clarity of illustration. A portion of a PCRAM array is shown. As will be described in greater detail below, the PCRAM cells **58** are formed in a checkerboard layout. Such an array of PCRAM cells **58** is formed in a self-aligned manner by etching a stack of conductive and phase change material (PCM) layers twice: first using a pattern of the bit lines **66**, and again using a pattern of the word lines **62**. The etching processes form the word lines **62**, the bit lines **66**, and the PCRAM cells **58**, with each PCRAM cell **58** being disposed at an intersection of a word line **62** and a bit line **66** in the top-down view.

(20) The bit lines **66** extend along a first direction D.sub.1, which is parallel to the active surface of the semiconductor substrate **70** (see FIG. **2**) and parallel to the longitudinal axes of the gates of the devices **72** (e.g., transistors). The bit lines **66** each emanate from a bit line pad **68**. Each bit line pad **68** is coupled to at least one bit line **66**. Although illustrated as separate elements, as will be described in greater detail below, each bit line pad **68** and its corresponding bit lines **66** are actually a single continuous conductive feature. The bit line pads **68** are connected to conductive features (e.g., interconnects) of an underlying metallization pattern (such as to the metallization layer **M4** in the example of FIG. **2**) by the conductive vias **92**. As will be described in greater detail below, the conductive vias **92** are electrically coupled to the bottom electrodes **82** of the PCRAM cells **58** (see FIG. **2**). As such, each conductive via **92** can also be referred to as a bottom electrode via (BEVA).

(21) The word lines **62** extend along a second direction D.sub.2, which is parallel to the active surface of the semiconductor substrate **70** (see FIG. **2**) and is perpendicular to the first direction D.sub.1 (e.g., perpendicular to the longitudinal axes of the gates of the devices **72** (e.g., transistors)). The word lines **62** each emanate from a word line pad **64**. Each word line pad **64** is coupled to at least one word line **62**. Although illustrated as separate elements, as will be described in greater detail below, each word line pad **64** and its corresponding word lines **62** are actually a single continuous conductive feature. The word line pads **64** are connected to conductive features (e.g., interconnects) of an overlying metallization pattern (such as to the metallization layer **M6** in the example of FIG. **2**) by the conductive vias **94**. As will be described in greater detail below, the conductive vias **94** are electrically coupled to the top electrodes **86** of the PCRAM cells **58** (see FIG. **2**). As such, each conductive via **94** can also be referred to as a top electrode via (TEVA).

(22) FIG. **21B** further illustrates several reference cross-sections. Cross-section **50C** is across several PCRAM cells **58**. Cross-section **50P.sub.1** is parallel to cross-section **50C**, and is across a bit line pad **68**. Cross-section **50P.sub.2** is perpendicular to cross-section **50C**, and is across a word line pad **64**. Subsequent figures refer to these cross-sections for clarity.

(23) FIGS. **3** through **21B** are various views of intermediate stages in the manufacturing of the semiconductor device **50**, in accordance with some embodiments. Specifically, the manufacturing of the interconnect structure **78** (see FIG. **2**) for the semiconductor device **50** is shown. As noted above, the interconnect structure **78** includes the PCRAM cells **58** of the PCRAM array **52** (see FIG. **1**).

(24) FIGS. **3**, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11A**, **12A**, **13**, **14**, **15**, **16**, **17A**, **18A**, **19**, **20**, and **21A** are cross-sectional view illustrating the logic region **50L** and the memory region **50M**, including a cell region **50C** (which illustrates cross-section **50C** in FIG. **21B**), a first pad region **50P.sub.1** (which illustrates cross-section **50P.sub.1** in FIG. **21B**), and a second pad region **50P.sub.2** (which illustrates cross-section **50P.sub.2** in FIG. **21B**). As will be described in greater detail below, a bit line pad **68** (see FIG. **12A**) will be formed in the first pad region **50P.sub.1**, a word line pad **64** (see FIG. **18A**) will be formed in the second pad region **50P.sub.2**, and the PCRAM cells **58** (see FIG. **18A**) will be formed in the cell region **50C**. Although the first pad region **50P.sub.1**, the second pad

region 50P.sub.2, and the cell region 50C are each illustrated in the same cross-sectional view, it should be appreciated that each of the regions are in different cross-sections, as shown by FIG. 21B.

(25) FIGS. 11B, 12B, 17B, 18B, and 21B are top-down views illustrating the memory region 50M. FIGS. 11B, 12B, 17B, 18B, and 21B show the semiconductor device 50 at a similar step of processing as FIGS. 11A, 12A, 17A, 18A, and 21A, respectively. FIGS. 11B, 12B, 17B, 18B, and 21B are simplified views, and some features are omitted for clarity of illustration.

(26) In FIG. 3, a metallization layer (e.g., M4, see FIG. 2) of the interconnect structure is formed. The metallization layer includes an IMD layer 102 and conductive features 104 (which can correspond to the metal lines L4, see FIG. 2). The IMD layer 102 is formed over the ILD layer(s) 74. The IMD layer 102 may be formed of any suitable dielectric material, for example, an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; a nitride such as silicon nitride; or the like. The IMD layer 102 may be formed by any suitable deposition process, such as spin coating, PVD, chemical vapor deposition (CVD), the like, or a combination thereof. The IMD layer 102 may be a layer formed of a low-k dielectric material having a k-value lower than about 3.0. The IMD layer 102 may be a layer formed of an extra-low-k (ELK) dielectric material having a k-value of less than 2.5.

(27) Conductive features 104 are formed in the IMD layer 102, and are electrically coupled to the devices 72. In accordance with some embodiments, the conductive features 104 include diffusion barrier layers and conductive material over the diffusion barrier layers. Openings are formed in the IMD layer 102 using, e.g., an etching process. The openings expose underlying conductive features, such as underlying metal vias. The diffusion barrier layers may be formed of tantalum nitride, tantalum, titanium nitride, titanium, cobalt-tungsten, or the like, and may be formed in the openings by a deposition process such as atomic layer deposition (ALD) or the like. The conductive material may include copper, aluminum, tungsten, silver, and combinations thereof, or the like, and may be formed over the diffusion barrier layers in the openings by an electro-chemical plating process, CVD, ALD, PVD, the like, or a combination thereof. In an embodiment, the conductive material is copper, and the diffusion barrier layers are thin barrier layers that prevent the copper from diffusing into the IMD layer 102. After formation of the diffusion barrier layers and the conductive material, excess of the diffusion barrier layers and conductive material may be removed by, for example, a planarization process such as a chemical mechanical polish (CMP) process. In some embodiments, the conductive features 104 are metal lines (which can correspond to the metal lines L4, see FIG. 2).

(28) An etch stop layer 106 is formed on the conductive features 104 and the IMD layer 102. The etch stop layer 106 may be formed of a dielectric material such as aluminum nitride, aluminum oxide, silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, a combination thereof, or the like. The etch stop layer 106 may be formed by chemical vapor deposition (CVD), PVD, ALD, a spin-on-dielectric process, the like, or a combination thereof. The etch stop layer 106 may also be a composite layer formed of a plurality of different dielectric sublayers. For example, the etch stop layer 106 may include a silicon carbide sublayer and an aluminum oxide sublayer formed on the silicon carbide sublayer. The silicon carbide sublayer can be used as a glue layer to improve adhesion between the aluminum oxide sublayer and the IMD layer 102.

(29) An IMD layer 108 is formed on the etch stop layer 106. In some embodiments, the IMD layer 108 is formed of a tetraethyl orthosilicate (TEOS) oxide (e.g., silicon oxide deposited using, e.g., a chemical vapor deposition (CVD) process with TEOS as a precursor). In some embodiments, the IMD layer 108 may be formed using PSG, BSG, BPSG, undoped silicate glass (USG), fluorosilicate glass (FSG), SiOCH, flowable oxide, a porous oxide, or the like, or combinations thereof. The IMD layer 108 may also be formed of a low-k dielectric material with a k value lower than about 3.0, for example. The IMD layer 108 can be formed to a thickness in the range of about

50 nm to about 150 nm.

(30) Via openings **110** are patterned in the IMD layer **108**. The via openings **110** may be formed using suitable photolithography and etching techniques. In some embodiments, an anti-reflective layer, such as a nitrogen-free anti-reflective coating (NFARC) (not shown) can be formed on the IMD layer **108** to protect the underlying layers during the patterning of via openings **110**.

(31) In FIG. **4**, conductive vias **92** are formed in the via openings **110**. The conductive vias **92** can also be referred to as BEVAs. In some embodiments, the conductive vias **92** include main conductive regions and conductive barrier layers lining sidewalls and bottom surfaces of the main conductive regions. The conductive barrier layers may be formed of titanium, titanium nitride, tantalum, tantalum nitride, cobalt, a combination thereof, or the like. The main conductive regions may be formed of metals such as copper, aluminum, tungsten, cobalt, alloys thereof, or the like. The formation of the conductive vias **92** may include conformally forming a conductive barrier layer extending into the via openings **110**, depositing a metallic material over the conductive barrier layer, and performing a planarization process, such as a CMP process or a mechanical grinding process, to remove excess portions of the conductive barrier layer and the metallic material from the top surface of the IMD layer **108**.

(32) In FIG. **5**, a plurality of memory cell layers are formed over the conductive vias **92** and the IMD layer **108**. Specifically, a bit line layer **114**, a bottom electrode layer **116**, a PCM layer **118**, and a top electrode layer **120** are deposited. The bit line layer **114** will be patterned in subsequent processing (see FIGS. **12A** and **12B**) to form bit lines **66** and bit line pads **68**. The top electrode layer **120**, the PCM layer **118**, and the bottom electrode layer **116** will also be patterned in subsequent processing (see FIGS. **18A** and **18B**) to form, respectively, the top electrodes **86**, the PCM elements **84**, and the bottom electrodes **82** of respective PCRAM cells **58**.

(33) The bit line layer **114** is formed on the conductive vias **92** and the IMD layer **108**. The bit line layer **114** is formed of a metal such as tungsten, titanium, cobalt, nickel, the like, or combinations thereof, and may be deposited by CVD, PVD, ALD, or the like. The bit line layer **114** is conformally formed, and may be formed using CVD, PVD, ALD, electro-chemical plating, electroless plating, or the like. In some embodiments, the bit line layer **114** is a layer of tungsten formed by CVD.

(34) The bottom electrode layer **116** is formed on the bit line layer **114**. The bottom electrode layer **116** is formed of a conductive material such as titanium, tantalum, aluminum, tungsten, platinum, nickel, chromium, ruthenium, nitrides thereof, combinations thereof, multilayers thereof, or the like. The bottom electrode layer **116** is conformally formed, and may be formed using CVD, PVD, ALD, electro-chemical plating, electroless plating, or the like. In some embodiments, the bottom electrode layer **116** is a layer of titanium nitride formed by PVD.

(35) The PCM layer **118** is formed on the bottom electrode layer **116**. The PCM layer **118** is formed of a chalcogenide material. Chalcogenide materials include at least a chalcogen anion (e.g., selenium (Se), tellurium (Te), and the like) and an electropositive element (e.g., germanium (Ge), silicon (Si), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), zinc (Zn), nitrogen (N), boron (B), carbon (C), and the like). An acceptable chalcogenide material includes, but is not limited to, $\text{GeSb}_{0.2}\text{Te}_{0.5}$ (GST). The PCM layer **118** is conformally formed, and may be formed using PVD, CVD, ALD, or the like. In some embodiments, the PCM layer **118** is a layer of GST formed by PVD. Forming the PCM layer **118** by PVD allows for a good film quality and may reduce gap-filling concerns.

(36) The top electrode layer **120** is formed on the PCM layer **118**. The top electrode layer **120** may be formed of a material that is selected from the same group of candidate materials of the bottom electrode layer **116**, and may be formed using a method that is selected from the same group of candidate methods for forming the bottom electrode layer **116**. The bottom electrode layer **116** and the top electrode layer **120** may be formed from the same material, or may include different materials.

(37) In FIGS. 6 through 12B, the bit line layer 114 is patterned to form bit lines 66 and bit line pads 68 (see FIGS. 12A and 12B). The top electrode layer 120, the PCM layer 118, and the bottom electrode layer 116 are also patterned to form top electrode strips 150, PCM strips 148, and bottom electrode strips 146 (see FIGS. 12A and 12B). This patterning process is the first of two patterning process performed to form the PCRAM cells 58 (see FIG. 2) in a self-aligned manner. In subsequent processing, the top electrode strips 150, the PCM strips 148, and the bottom electrode strips 146 will be patterned again to form the PCRAM cells 58.

(38) As will be discussed in greater detail below, FIGS. 6 through 12B illustrate a process in which a first mask 136 (see FIG. 10) is formed having a pattern of the bit lines 66 and a second mask 138 (see FIGS. 11A and 11B) is formed having a pattern of the bit line pads 68. In the illustrated embodiment, the first mask 136 is formed with a multiple-patterning process and the second mask 138 is formed with a single-patterning process, so that the features of the first mask 136 can be smaller than the features of the second mask 138. The bit line layer 114 is then patterned using both masks 136, 138 as a combined etching mask to simultaneously form the bit lines 66 and the bit line pads 68 (see FIGS. 12A and 12B).

(39) In FIG. 6, a plurality of masking layers are formed over the memory cell layers, e.g., over the top electrode layer 120. Specifically, one or more dielectric layers(s) 122 and a mandrel layer 124 are deposited. The dielectric layers(s) 122 will be patterned to form etching masks, which will be used in subsequent processing to pattern the bit line layer 114.

(40) The dielectric layers(s) 122 are formed on the top electrode layer 120. In the illustrated embodiment, the dielectric layers(s) 122 include a first dielectric layer 122A over the top electrode layer 120 and a second dielectric layer 122B over the first dielectric layer 122A. The first dielectric layer 122A may be a mask layer, such as a hard mask layer; may be formed of a nitride such as silicon nitride, silicon oxynitride, titanium nitride, or the like; and may be formed by deposition such as by PECVD, ALD, or the like. The second dielectric layer 122B may be a pad layer; may be formed of an oxide such as silicon oxide, a TEOS oxide, or the like; and may be formed by deposition such as by PECVD, ALD, or the like.

(41) The mandrel layer 124 is formed on the dielectric layers(s) 122, e.g., on the second dielectric layer 122B. The mandrel layer 124 is formed of a material that has a high etching selectivity from the etching of the underlying layer(s), e.g., the dielectric layers(s) 122. The mandrel layer 124 may be formed of a material such as amorphous silicon, polysilicon, silicon nitride, silicon oxide, the like, or combinations thereof, and may be formed using a process such as a CVD, PECVD, or the like.

(42) One or more masks are formed over the mandrel layer 124. The masks will be used to pattern the mandrel layer 124 and form mandrels. In some embodiments, the one or more masks may comprise one or more hard masks, a trilayer mask, a combination thereof, or the like. For example, a hard mask layer 126 can be formed over the mandrel layer 124 and a photosensitive mask 128 can be formed over the hard mask layer 126. In some embodiments, the hard mask layer 126 is formed of an oxide such as silicon oxynitride, silicon oxide, titanium oxide, a combination thereof, or the like. The photosensitive mask 128 may be a photoresist, such as a single-layer photoresist, a bilayer photoresist, a trilayer photoresist, or the like.

(43) In FIG. 7, the mandrel layer 124 is patterned to form mandrels 130. In the illustrated embodiment, the pattern of the photosensitive mask 128 is transferred to the hard mask layer 126, and the pattern of the hard mask layer 126 is then transferred to the mandrel layer 124. Each pattern may be transferred by an acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic. In some embodiment, the final etch is selective to the mandrel layer 124, e.g., selectively etches the material of the mandrel layer 124 at a faster rate than the material of the underlying dielectric layer 122, e.g., the second dielectric layer 122B. The photosensitive mask 128 and the hard mask layer 126 can optionally be removed with the material of the mandrel layer 124, or can be removed in a

subsequent cleaning process.

(44) After the patterning, the mandrels **130** can be separated by a spacing distance $D_{\text{sub.3}}$ in the range of about 40 nm to about 80 nm. Each of the mandrels **130** can have a width $W_{\text{sub.1}}$ in the range of about 40 nm to about 80 nm. The mandrels **130** will be used to pattern spacers over the dielectric layer(s) **122**. The spacing distance $D_{\text{sub.3}}$ and the width $W_{\text{sub.1}}$ of the mandrels **130** determines the spacing distance between the subsequently patterned spacers.

(45) In FIG. **8**, a spacer layer **132** is formed over the mandrels **130** and the dielectric layer(s) **122**. After formation, the spacer layer **132** extends along the top surfaces of the mandrels **130**, the sidewalls of the mandrels **130**, and the top surface of the underlying dielectric layer **122**, e.g., the second dielectric layer **122B**. The spacer layer **132** is formed of a material that has a high etching selectivity from the etching of the underlying layer(s), e.g., the dielectric layer(s) **122**. The spacer layer **132** may be formed from silicon nitride, aluminum oxide, aluminum nitride, tantalum nitride, titanium nitride, titanium oxide, the like, or combinations thereof, and may be formed using a process such as ALD, CVD, or the like. The spacer layer **132** has a high degree of conformality, with the thickness $T_{\text{sub.1}}$ of its vertical portions being equal to or slightly less than the thickness $T_{\text{sub.2}}$ of its horizontal portions. For example, the thickness $T_{\text{sub.1}}$ can be from about 80% to about 100% of the thickness $T_{\text{sub.2}}$. The thickness $T_{\text{sub.1}}$ can be in the range of about 15 nm to about 30 nm and the thickness $T_{\text{sub.2}}$ can be in the range of about 15 nm to about 30 nm. The spacer layer **132** will be patterned to form spacers over the dielectric layer(s) **122**. The thickness $T_{\text{sub.1}}$ of the vertical portions of the spacer layer **132** determines the width of the subsequently patterned spacers.

(46) In FIG. **9**, the spacer layer **132** is patterned to form spacers **134** over the dielectric layer(s) **122**. A suitable etching process is performed to remove the horizontal portions of the spacer layer **132**. The etching process selectively etches the horizontal portions of the spacer layer **132** at a faster rate than the mandrels **130** and the vertical portions of the spacer layer **132**. For example, when the spacer layer **132** is formed of silicon nitride, the etching process can be an anisotropic dry etch performed with methane ($\text{CH}_{\text{sub.4}}$), chlorine ($\text{Cl}_{\text{sub.2}}$), nitrogen ($\text{N}_{\text{sub.2}}$), or the like. After the etching process, the spacers **134** comprise the remaining vertical portions of the spacer layer **132**. The mandrels **130** can optionally be removed with the horizontal portions of the spacer layer **132**, or can be removed in a subsequent cleaning process. In some embodiments, the mandrels **130** are removed after the spacers **134** are formed, and can be removed by a suitable etching processes that selectively etches the mandrels **130** at a faster rate than the spacers **134**.

(47) After the patterning, the spacers **134** have a width $W_{\text{sub.2}}$ and are separated by a spacing distance $D_{\text{sub.4}}$. The spacing distance $D_{\text{sub.4}}$ between the spacers **134** can be in the range of about 20 nm to about 50 nm, and the width $W_{\text{sub.2}}$ of the spacers **134** can be in the range of about 15 nm to about 30 nm. As noted above, the spacing distance $D_{\text{sub.3}}$ and the width $W_{\text{sub.1}}$ (see FIG. **7**) of the mandrels **130** determines the spacing distance $D_{\text{sub.4}}$ between the spacers **134**, and the thickness $T_{\text{sub.1}}$ (see FIG. **8**) of the vertical portions of the spacer layer **132** determines the width $W_{\text{sub.2}}$ of the spacers **134**. Because selective etching process are used to form the spacers **134**, the thickness $T_{\text{sub.1}}$ of the vertical portions of the spacer layer **132** decreases by a small amount when forming the spacers **134**. The spacers **134** will be used to pattern the bit line layer **114**. The spacing distance $D_{\text{sub.4}}$ and the width $W_{\text{sub.2}}$ of the spacers **134** determines the spacing distance and the width of the resulting bit lines **66** (see FIGS. **12A** and **12B**).

(48) In FIG. **10**, the spacers **134** in undesired locations are removed in a cut process. The cut process may be performed using suitable photolithography and etching techniques. For example, a first subset of the spacers **134** may be covered with, e.g., a mask such as a photoresist, and an uncovered second subset of the spacers **134** may then be removed with an etch that selectively etches the material of the spacers **134** at a faster rate than the material of the underlying dielectric layer **122**, e.g., the second dielectric layer **122B**. In some embodiments, the spacers **134** are initially formed in both the logic region **50L** and memory region **50M**, and the cut process is used to remove

the spacers **134** from the logic region **50L**, the first pad region **50P.sub.1**, and the second pad region **50P.sub.2** so that the spacers **134** only remain in the cell region **50C**. The remaining spacers **134** form a first mask **136** in the cell region **50C**.

(49) In FIG. **11A**, a second mask **138** is formed in the first pad region **50P.sub.1**. The second mask **138** may be photosensitive mask, such as a photoresist, such as a single-layer photoresist, a bilayer photoresist, a trilayer photoresist, or the like. The second mask **138** is not formed by patterning spacers, and as such, the features of the second mask **138** are larger than the features of the first mask **136**. For example, the features of the second mask **138** can have a width $W_{sub.3}$ that is larger than the width $W_{sub.2}$ (see FIG. **9**). For example, the width $W_{sub.3}$ can be in the range of about 50 nm to about 500 nm.

(50) As shown by FIG. **11B**, portions of the first mask **136** and portions of the second mask **138** overlap. Thus, some of the patterned features will be continuous with one another. Further, as will be discussed in greater detail below, the bit lines **66** extend along the same direction $D_{sub.1}$ (see FIG. **12B**). The spacers **134**, which will be used to pattern the bit lines **66**, thus also extend along the same direction $D_{sub.1}$.

(51) In FIG. **12A**, the masks **136**, **138** are used as a combined etching mask to etch and pattern the dielectric layer(s) **122**. At least one of the dielectric layer(s) **122**, e.g., the first dielectric layer **122A**, remains after the etching and forms a patterned hard mask. The patterned hard mask is then used as an etching mask to etch and pattern the top electrode layer **120**, the PCM layer **118**, the bottom electrode layer **116**, and the bit line layer **114**. The patterning may include one or more etching processes. The etching method may include a plasma etching method, such as ion beam etching (IBE). IBE offers a high level of precision (e.g., high anisotropism), which can help control the profile of the resulting bit lines **66**. The etching may be implemented using glow discharge plasma (GDP), capacitive coupled plasma (CCP), inductively coupled plasma (ICP), or the like. The first mask **136**, the second mask **138**, and/or the dielectric layer(s) **122** may be consumed in the etching process, or may be removed after the etching process. In the illustrated embodiment, the first dielectric layer **122A** remains after the etching process.

(52) The etching process forms bit lines **66** and bit line pads **68**. The bit lines **66** and the bit line pads **68** are disposed beneath the top electrode strips **150**. The bit lines **66** and the bit line pads **68** comprise remaining portions of the bit line layer **114**.

(53) The etching process also forms top electrode strips **150**, PCM strips **148**, and bottom electrode strips **146**. The top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146** comprise remaining portions of the top electrode layer **120**, the PCM layer **118**, and the bottom electrode layer **116**, respectively. Although not shown, it should be appreciated that the patterned layers can have sloped sidewalls, and can have trapezoidal shapes in the illustrated cross-section. Each of the patterned layers has the same shape in the top-down view as the combined shape of the masks **136**, **138** (see FIG. **11B**).

(54) As shown by FIG. **12B**, each of the bit lines **66** are metal strips that extend along the same direction $D_{sub.1}$ and emanate from a bit line pad **68**. Thus, although the bit lines **66** and the bit line pads **68** are illustrated as separate elements, it should be appreciated that each bit line pad **68** and its corresponding bit lines **66** are actually a single continuous conductive feature that is patterned from the bit line layer **114**. In other words, the patterning shown in FIG. **12A** forms first conductive features that have bit line portions and bit line pad portions.

(55) It should be appreciated that FIGS. **6** through **12B** illustrate an example process for patterning the bit line layer **114**, and other processes may be used to pattern the bit line layer **114**. For example, the bit line layer **114** may also be patterned using next-generation lithography techniques such as extreme ultraviolet (EUV) lithography, deep ultraviolet (DUV) lithography, X-ray lithography, soft X-ray (SX) lithography, ion beam projection lithography, electron-beam projection lithography, or the like. The use of next-generation lithography techniques may allow the bit line layer **114** to be patterned by a single-patterning photolithography process, obviating the need for

multiple-patterning photolithography processes.

(56) In FIG. 13, spacers **156** are formed on the sidewalls of the bit line pads **68** and the bit lines **66**. The spacers **156** are also formed on the sidewalls of the top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146**. The spacers **156** may be formed by conformally depositing an insulating material and then etching the insulating material. The insulating material may be a nitride (e.g., silicon nitride, aluminum nitride, etc.), an oxide (e.g., silicon oxide, aluminum oxide, etc.), a carbide (e.g., silicon carbide), combinations thereof (e.g., silicon oxynitride, silicon carbonitride, etc.), multilayers thereof, or the like. The etch may be anisotropic.

(57) An IMD layer **158** is then formed over the spacers **156**, the IMD layer **108**, and the first dielectric layer **122A** (if present) or the top electrode strips **150**. The IMD layer **158** may be formed of a material that is selected from the same group of candidate materials of the IMD layer **108**, and may be formed using a method that is selected from the same group of candidate methods for forming the IMD layer **108**. The IMD layer **108** and the IMD layer **158** may be formed from the same material, or may include different materials.

(58) In FIG. 14, a planarization process is performed to remove excess material of the IMD layer **158**. The planarization process can be a CMP, mechanical grinding, an etchback, or the like. The planarization process removes the first dielectric layer **122A** (if present) to expose the top electrode strips **150**. After the planarization process, top surfaces of the IMD layer **158**, the spacers **156**, and the top electrode strips **150** are coplanar (within process variations).

(59) In FIG. 15, a word line layer **160** is formed on the planarized top surfaces of the IMD layer **158**, the spacers **156**, and the top electrode strips **150**. The word line layer **160** may be formed of a material that is selected from the same group of candidate materials of the bit line layer **114**, and may be formed using a method that is selected from the same group of candidate methods for forming the bit line layer **114**. The bit line layer **114** and the word line layer **160** may be formed from the same material, or may include different materials.

(60) In FIGS. 16 through 18B, the word line layer **160** is patterned to form word lines **62** and word line pads **64** (see FIGS. 18A and 18B). The top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146** are also patterned during the process for patterning the word line layer **160**, thus forming PCRAM cell **58** (see FIGS. 18A and 18B). This patterning process is the second of the two patterning process performed to form the PCRAM cells **58** (see FIGS. 18A and 18B) in a self-aligned manner.

(61) As will be discussed in greater detail below, FIGS. 16 through 18B illustrate a process in which a third mask **166** (see FIG. 16) is formed having a pattern of the word lines **62** and a fourth mask **168** (see FIGS. 17A and 17B) is formed having a pattern of the word line pads **64**. In the illustrated embodiment, the third mask **166** is formed with a multiple-patterning process and the fourth mask **168** is formed with a single-patterning process, so that the features of the third mask **166** can be smaller than the features of the fourth mask **168**. The word line layer **160** is then patterned using both masks **166**, **168** as a combined etching mask to simultaneously form the word lines **62** and the word line pads **64** (see FIGS. 18A and 18B).

(62) In FIG. 16, a plurality of masking layers are formed over the word line layer **160**. Specifically, one or more dielectric layers(s) **162** are deposited. The dielectric layers(s) **162** will be patterned to form etching masks, which will be used in subsequent processing to pattern the word line layer **160**.

(63) The dielectric layers(s) **162** are formed on the word line layer **160**. In the illustrated embodiment, the dielectric layers(s) **162** include a first dielectric layer **162A** over the word line layer **160** and a second dielectric layer **162B** over the first dielectric layer **162A**. The first dielectric layer **162A** may be a mask layer, such as a hard mask layer; may be formed of a nitride such as silicon nitride, silicon oxynitride, titanium nitride, or the like; and may be formed by deposition such as by PECVD, ALD, or the like. The second dielectric layer **162B** may be a pad layer; may be formed of an oxide such as silicon oxide, a TEOS oxide, or the like; and may be formed by

deposition such as by PECVD, ALD, or the like.

(64) Spacers **164** are then formed over the dielectric layer(s) **162**. The spacers **164** may be formed of a material that is selected from the same group of candidate materials of the spacers **134**, and may be formed using a method that is selected from the same group of candidate methods for forming the spacers **134**. For example, the spacers **164** can be formed by depositing a mandrel layer (see, e.g., FIG. **6**), patterning the mandrel layer to form mandrels (see, e.g., FIG. **7**), depositing a spacer layer over the mandrels (see, e.g., FIG. **8**), patterning the spacer layer to form the spacers **164** (see, e.g., FIG. **9**), and removing spacers **164** in undesired locations in a cut process (see, e.g., FIG. **10**). The spacers **134** and the spacers **164** may be formed from the same material, or may include different materials. The remaining spacers **164** form a third mask **166** in the cell region **50C**, which will be used to pattern the word line layer **160**.

(65) In FIG. **17A**, a fourth mask **168** is formed in the second pad region **50P.sub.2**. The fourth mask **168** may be photosensitive mask, such as a photoresist, such as a single-layer photoresist, a bilayer photoresist, a trilayer photoresist, or the like. The fourth mask **168** is not formed by patterning spacers, and as such, the features of the fourth mask **168** are larger than the features of the third mask **166**. In some embodiments, the features of the third mask **166** and the fourth mask **168** have similar dimensions as the features of the first mask **136** and the second mask **138**, respectively. In some embodiments, the features of the third mask **166** and the fourth mask **168** have different dimensions than the features of the first mask **136** and the second mask **138**.

(66) As shown by FIG. **17B**, portions of the third mask **166** and portions of the fourth mask **168** overlap. Thus, some of the patterned features will be continuous with one another. Further, as will be discussed in greater detail below, the word lines **62** extend along the same direction D.sub.2 (see FIG. **18B**). The spacers **164**, which will be used to pattern the word lines **62**, thus also extend along the same direction D.sub.2.

(67) In FIG. **18A**, the masks **166**, **168** are used as a combined etching mask to etch and pattern the dielectric layer(s) **162**. At least one of the dielectric layer(s) **162**, e.g., the first dielectric layer **162A**, remains after the etching and forms a patterned hard mask. The patterned hard mask is then used as an etching mask to etch and pattern the word line layer **160**, the top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146**. The patterning may include one or more etching processes, and can form recesses **170** in the IMD layer **158**. The etching method may include a plasma etching method, such as ion beam etching (IBE). IBE offers a high level of precision (e.g., high anisotropism), which can help control the profile of the resulting word lines **62**. The etching may be implemented using glow discharge plasma (GDP), capacitive coupled plasma (CCP), inductively coupled plasma (ICP), or the like. The third mask **166**, the fourth mask **168**, and/or the dielectric layer(s) **162** may be consumed in the etching process, or may be removed after the etching process. In the illustrated embodiment, the first dielectric layer **162A** remains after the etching process.

(68) The etching process forms word lines **62** and word line pads **64**. The word lines **62** are disposed over top electrodes **86** and the word line pads **64** are disposed over unpatterned portions of the IMD layer **158**. The word lines **62** and the word line pads **64** comprise remaining portions of the word line layer **160**. Although not shown, it should be appreciated that the word lines **62** can have sloped sidewalls, and can have trapezoidal shapes in the illustrated cross-section. The patterned word line layer **160** has the same shape in the top-down view as the combined shape of the masks **166**, **168** (see FIG. **17B**).

(69) As shown by FIG. **18B**, each of the word lines **62** are metal strips that extend along the same direction D.sub.2 and emanate from a word line pad **64**. Thus, although the word lines **62** and the word line pads **64** are illustrated as separate elements, it should be appreciated that each word line pad **64** and its corresponding word lines **62** are actually a single continuous conductive feature that is patterned from the word line layer **160**. In other words, the patterning shown in FIG. **18A** forms second conductive features that have word line portions and word line pad portions.

(70) The etching process also patterns the top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146** to form the bottom electrodes **82**, the PCM elements **84**, and the top electrodes **86**, respectively, which together form the PCRAM cells **58**. Each PCRAM cell **58** includes a bottom electrode **82**, a PCM element **84**, and a top electrode **86**, with the PCM element **84** being disposed between the bottom electrode **82** and the top electrode **86**. The bottom electrodes **82** comprise remaining portions of the bottom electrode strips **146**. The PCM elements **84** comprise remaining portions of the PCM strips **148**. The top electrodes **86** comprise remaining portions of the top electrode strips **150**. Although not shown, it should be appreciated that the bottom electrodes **82**, the PCM elements **84**, and the top electrodes **86** can have sloped sidewalls, and can have trapezoidal shapes in the illustrated cross-section.

(71) The etching process removes the portions of the top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146** that are uncovered by the masks **166**, **168**, such as the portions of those layers over the bit line pads **68**. Thus, as shown by FIG. **18B**, each PCRAM cell **58** is disposed at an intersection of a word line **62** and a bit line **66** in the top-down view. The PCRAM cells **58** are thus formed in a self-aligned manner, which allows the spacing distance $D_{sub.5}$ and the width $W_{sub.4}$ of the PCRAM cells **58** to be small. For example, the spacing distance $D_{sub.5}$ can be in the range of about 20 nm to about 50 nm, and the width $W_{sub.4}$ can be in the range of about 15 nm to about 30 nm. The spacing distance $D_{sub.5}$ corresponds to both the distance between adjacent PCRAM cells **58**, and also the distance between adjacent intersections of the word lines **62** and the bit lines **66**.

(72) The recesses **170** are formed in portions of the IMD layer **158**. Specifically, the recesses **170** are formed by etching the IMD layer **158** and any top electrode strips **150**, PCM strips **148**, or bottom electrode strips **146** that are uncovered by the masks **166**, **168**. Thus, the recesses **170** expose the bit line pads **68**. Timed etch processes may be used to stop the etching of the recesses **170** after the recesses **170** reach a desired depth. Although not shown in FIG. **18A**, it should be appreciated that the recesses **170** also expose portions of the bit lines **66** that are not disposed beneath the word lines **62** and the word line pads **64**. The recesses **170** can be formed by performing the etching process (described above) with multiple etches. For example, the etching process can include a first etch and a second etch. The first etch can selectively etch the material of the word line layer **160** at a faster rate than the materials of the IMD layer **158**, the top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146**. The second etch can selectively etch the materials of the IMD layer **158**, the top electrode strips **150**, the PCM strips **148**, and the bottom electrode strips **146** at a faster rate than the material of the bit lines **66** and the bit line pad **68**. In some embodiments, the first etch is ion beam etching using sulfur hexafluoride ($SF_{sub.6}$), argon (Ar), oxygen ($O_{sub.2}$), and difluoromethane ($CH_{sub.2}F_{sub.2}$) for a duration in a range of 20 seconds to 60 seconds, and the second etch is ion beam etching using chlorine ($Cl_{sub.2}$), hydrogen bromide (HBr), argon (Ar), and difluoromethane ($CH_{sub.2}F_{sub.2}$) for a duration in a range of 15 seconds to 75 seconds. Other etching parameters may be used when manufacturing magnetoresistive random-access memories (MRAMs), resistive random access memories (RRAMs), memories with a selector structure, or the like. Further, the etching parameters can be varied based on the materials of the etched films and the film thicknesses.

(73) After the recesses **170** are formed, the IMD layer **158** includes recessed portions **158R** and unrecessed portions **158U**. The recessed portions **158R** surround bit lines **66** and the bit line pads **68**, and the unrecessed portions **158U** surround the PCRAM cells **58**. The recessed portions **158R** are disposed beneath the word lines **62** and the word line pads **64**. The recessed portions **158R** are unpatterned and thus have a greater height than the unrecessed portions **158U**.

(74) It should be appreciated that FIGS. **16** through **18B** illustrate an example process for patterning the word line layer **160**, and other processes may be used to pattern the word line layer **160**. For example, the word line layer **160** may also be patterned using next-generation lithography techniques such as extreme ultraviolet (EUV) lithography, deep ultraviolet (DUV) lithography, X-

ray lithography, soft X-ray (SX) lithography, ion beam projection lithography, electron-beam projection lithography, or the like. The use of next-generation lithography techniques may allow the word line layer **160** to be patterned by a single-patterning photolithography process, obviating the need for multiple-patterning photolithography processes.

(75) In FIG. **19**, an IMD layer **178** is formed over the IMD layer **158**, the bit line pads **68**, and the first dielectric layer **162A** (if present) or the word lines **62** and the word line pads **64**. The IMD layer **178** may be formed of a material that is selected from the same group of candidate materials of the IMD layer **108**, and may be formed using a method that is selected from the same group of candidate methods for forming the IMD layer **108**. The IMD layer **108** and the IMD layer **178** may be formed from the same material, or may include different materials. After formation, the IMD layer **178** is disposed over the bit lines **66** and the recessed portions **158R** of the IMD layer **158**.

The IMD layer **178** thus surrounds the word lines **62**, the word line pads **64**, and the unrecessed portions **158U** of the IMD layer **158**. Further, the combination of the IMD layers **158**, **178** surround the PCRAM cells **58** on all four sides. Confining all four sides of the PCRAM cells **58** in a dielectric material can improve the performance of the PCRAM cells **58** during operation, as the dielectric material helps absorb heat that is generated when the PCM elements **84** change phases.

(76) In FIG. **20**, a planarization process is performed to remove excess material of the IMD layer **178**. The planarization process can be a CMP, mechanical grinding, an etchback, or the like. The planarization process removes the first dielectric layer **162A** (if present) to expose the word lines **62** and the word line pads **64**. After the planarization process, top surfaces of the IMD layer **178**, the word line pads **64**, and the word lines **62** are coplanar (within process variations).

(77) In FIG. **21A**, an etch stop layer **182** is formed on the planarized top surfaces of the IMD layer **178**, the word line pads **64**, and the word lines **62**. The etch stop layer **182** may be formed of a material that is selected from the same group of candidate materials of the etch stop layer **106**, and may be formed using a method that is selected from the same group of candidate methods for forming the etch stop layer **106**. The etch stop layer **106** and the etch stop layer **182** may be formed from the same material, or may include different materials.

(78) An IMD layer **184** is then formed on the etch stop layer **182**. The IMD layer **184** may be formed of a material that is selected from the same group of candidate materials of the IMD layer **108**, and may be formed using a method that is selected from the same group of candidate methods for forming the IMD layer **108**. The IMD layer **108** and the IMD layer **184** may be formed from the same material, or may include different materials.

(79) Conductive features **186** (e.g., interconnects) are then formed extending through the IMD layer **184** and the etch stop layer **182**. The conductive features **186** include conductive vias **186V** (which can correspond to the metal vias **V5**, see FIG. **2**, and to the conductive vias **94**, see FIG. **1**) and conductive lines **186L** (which can correspond to the metal lines **L5**, see FIG. **2**). The conductive features **186** are formed in both the memory region **50M** and the logic region **50L**. The conductive features **186** may be formed by a damascene process, such as a single damascene process, a dual damascene process, or the like. The conductive features **186** are electrically coupled to the memory devices (e.g., PCRAMs) formed in the memory region **50M** and the logic devices (e.g., logic circuits) formed in the logic region **50L**. A first subset of the conductive features **186A** are formed in the memory region **50M**, and are connected to the word line pads **64**. A second subset of the conductive features **186B** are formed in the logic region **50L**, and further extend through the IMD layer **178**, the IMD layer **158**, the IMD layer **108**, and the etch stop layer **106** to be connected to the conductive features **104**. In some embodiments, the conductive features **186** electrically couple the memory devices to the logic devices. For example, the conductive features **186** can be used to electrically couple some of the conductive features **104** to some of the word line pads **64**, such as in the illustrated metallization layer, or in another metallization layer. Although each conductive via **186V** and corresponding conductive line **186L** is illustrated as a separate element, it should be appreciated that they may be a continuous conductive feature, such as in embodiments where they

are formed by a dual damascene process.

(80) Each PCRAM cell **58** is connected to a conductive feature **186** and a conductive feature **104**. Specifically, each top electrode **86** is connected to a conductive line **186L** by a word line **62**, a word line pad **64**, and a conductive via **94**. Likewise, each bottom electrode **82** is connected to a conductive feature **104** by a bit line **66**, a bit line pad **68**, and a conductive via **92**. The word line pads **64** are disposed under and are physically and electrically coupled to the conductive vias **94**. The bit line pads **68** are disposed over and are physically and electrically coupled to the conductive vias **92**. Thus, the conductive vias **92** connect the bit line pads **68** to conductive features (e.g., interconnects) of an underlying metallization pattern (e.g., metallization layer **M4** in FIG. 2), and the conductive vias **94** connect the word line pads **64** to conductive features (e.g., interconnects) of an overlying metallization pattern (e.g., metallization layer **M6** in FIG. 2). Further, the conductive features **186B** connect the conductive features of the underlying metallization pattern to the conductive features of the overlying metallization pattern.

(81) As shown by FIG. 21B, the bit lines **66** emanating from each bit line pad **68** can be interleaved. For example, the bit lines **66** emanating from a first bit line pad **68A** can be interleaved with the bit lines **66** emanating from a second bit line pad **68B**. Similarly, the word lines **62** emanating from each word line pad **64** can be interleaved. For example, the word lines **62** emanating from a first word line pad **64A** can be interleaved with the word lines **62** emanating from a second word line pad **64B**.

(82) FIG. 22 is a cross-sectional view of the semiconductor device **50**, in accordance with some other embodiments. This embodiment is similar to the embodiment described with respect to FIG. 21A, except ovonic threshold switching (OTS) layers **192** are formed between the bottom electrodes **82** and the bit lines **66**. The OTS layers **192** may be formed of a chalcogenide material, and can be formed of a different chalcogenide material than the PCM elements **84**. The OTS layers **192** may be used for memory selectors. Memory selectors function similar to transistors, having a threshold voltage ($V_{sub.th}$) value. Only applied voltages larger than the $V_{sub.th}$ of a PCRAM cell **58** can create a current path to the PCRAM cell **58**, thus enabling memory read/write operations.

(83) FIGS. 23A through 23E are three-dimensional views of intermediate stages in a self-aligned patterning process for forming the PCRAM cells, in accordance with some embodiments. FIGS. 23A through 23E are simplified views, and some features are omitted for clarity of illustration. FIGS. 23A through 23E illustrate additional views of the semiconductor device **50** during the process described above with respect to FIGS. 3 through 21B.

(84) In FIG. 23A, the bit line layer **114** is deposited over a substrate, and the memory cell layers (e.g., the bottom electrode layer **116**, the PCM layer **118**, and the top electrode layer **120**) are deposited over the bit line layer **114**. The bit line layer **114** and the memory cell layers are deposited in the manner described with respect to FIG. 5.

(85) In FIG. 23B, the bottom electrode layer **116**, the PCM layer **118**, the top electrode layer **120**, and the bit line layer **114** are patterned. Patterning the bit line layer **114** forms the bit lines **66** and the bit line pads **68**. Patterning the bottom electrode layer **116**, the PCM layer **118**, and the top electrode layer **120** forms the bottom electrode strips **146**, the PCM strips **148**, and the top electrode strips **150**, respectively. The bottom electrode strips **146**, the PCM strips **148**, the top electrode strips **150**, and the bit lines **66** each extend in the first direction $D_{sub.1}$. The bottom electrode layer **116**, the PCM layer **118**, the top electrode layer **120**, and the bit line layer **114** are patterned in the manner described with respect to FIGS. 6 through 12B.

(86) In FIG. 23C, the IMD layer **158** is deposited around the bottom electrode strips **146**, the PCM strips **148**, the top electrode strips **150**, the bit lines **66**, and the bit line pads **68**. The IMD layer **158** is then planarized to expose the top electrode strips **150**. The IMD layer **158** is deposited and planarized in the manner described with respect to FIGS. 13 and 14.

(87) In FIG. 23D, the word line layer **160** is deposited over the IMD layer **158**, the top electrode strips **150**, and the other underlying strips. The word line layer **160** is deposited in the manner

described with respect to FIG. 15.

(88) In FIG. 23E, the word line layer **160**, the bottom electrode strips **146**, the PCM strips **148**, and the top electrode strips **150** are patterned. Patterning the word line layer **160** forms the word lines **62** and the word line pads **64**. Patterning the bottom electrode strips **146**, the PCM strips **148**, and the top electrode strips **150** forms the PCRAM cells **58**. The word lines **62** each extend in the second direction D.sub.2, which is perpendicular to the first direction D.sub.1. As noted above, the PCRAM cells **58** are each disposed at an intersection of a word line **62** and a bit line **66** in a top-down view. The word line layer **160**, the bottom electrode strips **146**, the PCM strips **148**, and the top electrode strips **150** are patterned in the manner described with respect to FIGS. 16 through 18B.

(89) Embodiments may achieve advantages. Forming the PCRAM cells **58** in a self-aligned manner with multiple patterning processes allows the PCRAM cells **58** to be formed with smaller spacing distances and widths, thereby improving the density and the performance of the PCRAM cells **58**. Specifically, smaller PCRAM cells **58** generate less heat when their PCM elements **84** change phases. Further, embodiments patterning processes allow the word lines **62** and the bit lines **66** for the PCRAM array to be simultaneously patterned with the PCRAM cells **58**, allowing manufacturing costs to be reduced.

(90) In an embodiment, a device includes: a first metallization layer over a substrate, the substrate including active devices; a first bit line over the first metallization layer, the first bit line connected to first interconnects of the first metallization layer, the first bit line extending in a first direction, the first direction parallel to gates of the active devices; a first phase-change random access memory (PCRAM) cell over the first bit line; a word line over the first PCRAM cell, the word line extending in a second direction, the second direction perpendicular to the gates of the active devices; and a second metallization layer over the word line, the word line connected to second interconnects of the second metallization layer.

(91) In some embodiments, the device further includes: a second bit line over the first metallization layer, the second bit line extending in the first direction; and a second PCRAM cell over the second bit line, the word line disposed over the second PCRAM cell, the first PCRAM cell being separated from the second PCRAM cell by a distance in a range of 20 nm to 50 nm. In some embodiments of the device, the first PCRAM cell and the second PCRAM cell each have a width in a range of 15 nm to 30 nm. In some embodiments, the device further includes: a bit line pad over the first metallization layer, the bit line pad and the first bit line being a first continuous conductive feature; a first conductive via connecting the bit line pad to the first interconnects of the first metallization layer; and a first inter-metal dielectric (IMD) layer around the first conductive via, the first continuous conductive feature disposed on the first IMD layer. In some embodiments, the device further includes: a word line pad over the bit line pad, the word line pad and the word line being a second continuous conductive feature; a second conductive via connecting the word line pad to the second interconnects of the second metallization layer; and a second IMD layer around the second conductive via, the second IMD layer disposed on the second continuous conductive feature. In some embodiments, the device further includes: a third conductive via extending through the first IMD layer and the second IMD layer, the third conductive via connecting the first interconnects of the first metallization layer to the second interconnects of the second metallization layer. In some embodiments, the device further includes: a first dielectric layer having a first portion and a second portion, the first portion surrounding the first bit line, the second portion surrounding the first PCRAM cell, the first portion having a first height, the second portion having a second height, the second height being greater than the first height, the word line disposed over the second portion. In some embodiments, the device further includes: a second dielectric layer over the first bit line and the first portion of the first dielectric layer, the second dielectric layer surrounding the word line and the second portion of the first dielectric layer. In some embodiments of the device, the first PCRAM cell includes: a bottom electrode connected to the first bit line; a top electrode connected

to the word line; and a phase change material (PCM) element between the top electrode and the bottom electrode. In some embodiments, the device further includes: an ovonic threshold switching layer between the bottom electrode and the first bit line.

(92) In an embodiment, a device includes: a first inter-metal dielectric (IMD) layer; a first conductive via extending through the first IMD layer; a first conductive feature having a bit line pad portion and a bit line portion, the bit line pad portion disposed on the first conductive via, the bit line portion disposed on the first IMD layer; a phase-change random access memory (PCRAM) cell on the bit line portion of the first conductive feature; a second IMD layer surrounding the PCRAM cell and the first conductive feature; and a second conductive feature having a word line pad portion and a word line portion, the word line pad portion disposed on the second IMD layer, the word line portion disposed on the PCRAM cell.

(93) In some embodiments, the device further includes: a second conductive via on the word line pad portion of the second conductive feature; and a third IMD layer surrounding the second conductive via.

(94) In an embodiment, a method includes: forming a bit line layer over a substrate including active devices; depositing a phase change material (PCM) layer over the bit line layer; patterning the PCM layer and the bit line layer to form a PCM strip and a bit line, respectively, the PCM strip and the bit line each extending in a first direction in a top-down view, the first direction parallel to gates of the active devices; depositing a first inter-metal dielectric (IMD) layer around the PCM strip and the bit line; depositing a word line layer over the first IMD layer and the PCM strip; and patterning the word line layer and the PCM strip to form a word line and a PCM element, respectively, the word line extending in a second direction in the top-down view, the second direction perpendicular to the gates of the active devices, the PCM element disposed at an intersection of the word line and the bit line in the top-down view.

(95) In some embodiments of the method, patterning the PCM layer and the bit line layer includes: forming a first mask over the PCM layer, features of the first mask extending in the first direction, the features of the first mask having a first width; forming a second mask over the PCM layer, features of the second mask extending in the second direction, the features of the second mask having a second width, portions of the first mask and the second mask overlapping; and etching the PCM layer and the bit line layer using the first mask and the second mask as a first combined etching mask to form a first conductive feature, the first conductive feature including the bit line and a bit line pad. In some embodiments of the method, patterning the word line layer and the PCM strip includes: forming a third mask over the word line layer, features of the third mask having the first width; forming a fourth mask over the PCM layer, features of the fourth mask having the second width, portions of the third mask and the fourth mask overlapping; and etching the word line layer and the PCM strip using the third mask and the fourth mask as a second combined etching mask to form a second conductive feature, the second conductive feature including the word line and a word line pad. In some embodiments of the method, etching the word line layer and the PCM strip includes: etching the word line layer with ion beam etching using hexafluoride (SF₆), argon (Ar), oxygen (O₂), and difluoromethane (CH₂F₂) for a duration in a range of 20 seconds to 60 seconds; and etching the PCM strip and the first IMD layer with ion beam etching using chlorine (Cl₂), hydrogen bromide (HBr), argon (Ar), and difluoromethane (CH₂F₂) for a duration in a range of 15 seconds to 75 seconds. In some embodiments, the method further includes: depositing the bit line layer on a first conductive via, the bit line pad contacting the first conductive via; and forming a second conductive via contacting the word line pad. In some embodiments of the method, patterning the word line layer and the PCM strip includes exposing a portion of the bit line layer covered by the PCM strip. In some embodiments of the method, patterning the word line layer and the PCM strip includes recessing a portion of the first IMD layer, the word line disposed on an unrecessed portion of the first IMD layer. In some embodiments, the method further includes: depositing a second IMD layer around the word line

and the unrecessed portion of the first IMD layer.

(96) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a first conductive feature having a bit line pad portion and a bit line portion, the bit line pad portion being larger than the bit line portion, the bit line portion emanating from the bit line pad portion; a phase-change random access memory cell over the bit line portion of the first conductive feature; an inter-metal dielectric around the phase-change random access memory cell and the first conductive feature; and a second conductive feature having a word line pad portion and a word line portion, the word line pad portion disposed over the inter-metal dielectric, the word line portion disposed over the phase-change random access memory cell, the word line pad portion being larger than the word line portion, the word line portion emanating from the word line pad portion.
2. The device of claim 1, wherein the phase-change random access memory cell comprises: a bottom electrode interfacing with the bit line portion; a top electrode interfacing with the word line portion; and a phase-change material element disposed between the bottom electrode and the top electrode.
3. The device of claim 1, further comprising: an ovonic threshold switching layer interfacing with the phase-change random access memory cell and the bit line portion.
4. The device of claim 3, wherein the phase-change random access memory cell comprises a phase-change material element, and the ovonic threshold switching layer comprises a different chalcogenide material than the phase-change material element.
5. The device of claim 1, further comprising: a spacer contacting a sidewall of the phase-change random access memory cell and a sidewall of the first conductive feature.
6. The device of claim 1, wherein the inter-metal dielectric electrically isolates the first conductive feature from the second conductive feature.
7. The device of claim 1, wherein the phase-change random access memory cell is disposed at a location where the word line portion crosses over the bit line portion.
8. The device of claim 1, further comprising: a first interconnect under the bit line pad portion; and a second interconnect over the word line pad portion.
9. The device of claim 1, wherein the inter-metal dielectric has a first portion and a second portion, the first portion disposed around the bit line pad portion, the second portion disposed around the bit line portion and the phase-change random access memory cell, the first portion having a first height, the second portion having a second height, the second height being greater than the first height.
10. A device comprising: a first metallization layer comprising a first interconnect; a second metallization layer comprising a second interconnect; and a third metallization layer between the first metallization layer and the second metallization layer, the third metallization layer comprising: a first conductive feature, the first conductive feature having a bit line pad portion and a bit line portion, the bit line pad portion connected to the first interconnect of the first metallization layer, the bit line portion extending in a first direction in a top-down view; a phase-change random access memory cell over the bit line portion; an inter-metal dielectric around the phase-change random

access memory cell and the first conductive feature; and a second conductive feature over the inter-metal dielectric, the second conductive feature having a word line pad portion and a word line portion, the word line pad portion connected to the second interconnect of the second metallization layer, the word line portion extending in a second direction in the top-down view, the second direction perpendicular to the first direction, the phase-change random access memory cell disposed at an intersection of the word line portion and the bit line portion in the top-down view.

11. The device of claim 10, wherein the inter-metal dielectric has a first portion adjacent the bit line portion and has a second portion adjacent the bit line pad portion, the first portion having a first height, the second portion having a second height, the second height being less than the first height.

12. The device of claim 10, wherein the phase-change random access memory cell comprises: a bottom electrode connected to the bit line portion; a top electrode connected to the word line portion; and a phase-change material element between the top electrode and the bottom electrode.

13. The device of claim 12, further comprising: a threshold switching layer between the bottom electrode and the bit line portion, the threshold switching layer comprising a first chalcogenide material, the phase-change material element comprising a second chalcogenide material, the second chalcogenide material different than the first chalcogenide material.

14. The device of claim 10, wherein the first conductive feature and the second conductive feature comprise the same metal.

15. The device of claim 10, wherein the bit line pad portion is wider than the bit line portion in the top-down view, and the word line pad portion is wider than the word line portion in the top-down view.

16. The device of claim 10, wherein the bit line portion is longer than the bit line pad portion in the top-down view, and the word line portion is longer than the word line pad portion in the top-down view.

17. A device comprising: a first metallization layer comprising a first interconnect; a second metallization layer comprising a second interconnect; and a third metallization layer between the first metallization layer and the second metallization layer, the third metallization layer comprising: a first conductive pad connected to the first interconnect of the first metallization layer; a first conductive line emanating from the first conductive pad in a first direction in a top-down view, the first conductive pad being larger than the first conductive line; a second conductive pad connected to the second interconnect of the second metallization layer; a second conductive line emanating from the second conductive pad in a second direction in the top-down view, the second conductive pad being larger than the second conductive line; and a phase-change random access memory cell at an intersection of the second conductive line and the first conductive line in the top-down view.

18. The device of claim 17, wherein the phase-change random access memory cell comprises: a bottom electrode connected to the first conductive line; a top electrode connected to the second conductive line; and a phase-change material element disposed between the bottom electrode and the top electrode.

19. The device of claim 17, wherein the first conductive pad and the first conductive line are part of a first continuous conductive feature, and the second conductive pad and the second conductive line are part of a second continuous conductive feature.

20. The device of claim 17, further comprising: an inter-metal dielectric around the first conductive pad and the first conductive line, the second conductive pad and the second conductive line being disposed over the inter-metal dielectric.
